# Algebraic methods



**1(a)** Simplify $3x^2 \times 5x^5$.

1(a)

**1(b)** Simplify $\frac{5x^3y^2}{15x^2y^3}$.

1(b)

**2(a)** Factorise $x^2 - 2x - 24$.

2(a)

**2(b)** Factorise $3x^2 - 17x + 20$.

2(b)

**3(a)** Calculate $197041 \div 23$ using long division.

3(a)

**3(b)** Calculate $56168 \div 34$ using long division.

3(b)

**4(a)** Find the line through $(-1,4)$ and $(5,-14)$.

4(a)

**4(b)** Find the line through $(2,-6)$ and $(8,-3)$.

4(b)

**5(a)** Complete the square: $x^2 - 2x - 20$.

5(a)

**5(b)** Complete the square: $2x^2 + 4x + 15$.

5(b)

## Chapter Overview

**Luisa · 3**

1. Algebraic fractions

Factorise and cancel common factors.

2. Dividing polynomials

Use long division, including remainders.

3. The factor theorem

Identify factors and determine unknown coefficients.

4. Proof

Deduction, exhaustion and counterexamples.

## 1: Simplifying Algebraic Fractions

**Luisa · 4**

Factorise first, then cancel common factors. Never cancel terms joined by addition or subtraction. Retain the restrictions from the original denominator.

**1**

$$\frac{x^2-1}{x^2+x}$$

1

**2**

$$\frac{x^2+3x+2}{x+1}$$

2

**3**

$$\frac{2x^2+11x+12}{x^2+9x+20}$$

3

**4**

$$\frac{4-x^2}{x^2+2x-8}$$

4

Tips: factorise the easier expression first. Also $\frac{a-b}{b-a}=-1$. Do not leave 1 as a denominator.

## Exercise 7A – Y1 Pages 138-139 **Luisa · 5**

2. Simplify as far as possible.

**m**

$$\frac{2x^2+3x+1}{x^2-x-2}$$

m

**n**

$$\frac{x^2+6x+8}{3x^2+7x+2}$$

n

**o**

$$\frac{2x^2-5x-3}{2x^2-9x+9}$$

o

**3** Find a,b,c given

$$\frac{6x^3+3x^2-84x}{6x^2-33x+42}=\frac{ax(x+b)}{x+c}$$

3

## Terminology **Luisa · 6**

$11 \div 4 = 2$ remainder 3

**Dividend** The number being divided

Dividend

**Divisor** The number we divide by

Divisor

**Quotient** The whole-number result

Quotient

**Remainder** The amount left over

Remainder

Dividend = divisor × quotient + remainder

## Normal Long Division — Luisa · 7

Calculate $423 \div 11$ using long division.

**1** Divide 42 by 11.

1

**2** Bring down the final digit, 3.

2

**3** State the quotient and remainder.

3

**4** Continue to a decimal.

4

Repeat: divide, multiply by the divisor, subtract, bring down the next digit.

## Example — Luisa · 8

Divide

$$\frac{6x^3 + 28x^2 - 7x + 15}{x + 5}$$

**1**

$6x^3 \div x = 6x^2$

1

**2**

$-2x^2 \div x = -2x$

2

**3**

$3x \div x = 3$

3

**Result** Combine the quotient terms and remainder.

Result

Include zero coefficients for missing powers. Subtract the whole product, including its signs; check by multiplying back.

## Further Example — Luisa · 9

Divide

$$\frac{3x^3 + 0x^2 - 2x + 4}{x - 1}$$

**1**

$3x^3 \div x = 3x^2$

1

**2**

$3x^2 \div x = 3x$

2

**3**

$x \div x = 1$

3

**Result** Combine the quotient terms and remainder.

Result

Include zero coefficients for missing powers. Subtract the whole product, including its signs; check by multiplying back.

## Test Your Understanding — Luisa · 10

Divide

$$\frac{2x^3 - 5x^2 - 16x + 10}{x - 4}$$

**1**

$2x^3 \div x = 2x^2$

1

**2**

$3x^2 \div x = 3x$

2

**3**

$-4x \div x = -4$

3

**Result** Combine the quotient terms and remainder.

Result

Include zero coefficients for missing powers. Subtract the whole product, including its signs; check by multiplying back.

## Further Test Your Understanding — Luisa · 11

Divide

$$\frac{8x^3 + 0x^2 + 0x - 1}{2x - 1}$$

**1**

$8x^3 \div 2x = 4x^2$

1

**2**

$4x^2 \div 2x = 2x$

2

**3**

$2x \div 2x = 1$

3

**Result** Combine the quotient terms and remainder.

Result

Include zero coefficients for missing powers. Subtract the whole product, including its signs; check by multiplying back.

Extension: $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$.

## Exercise 7B – Y1 Pages 142 — Luisa · 12

15 $\mathrm{f}(x) = 10x^3 + 43x^2 - 2x - 10$.

Find the remainder when $\mathrm{f}(x)$ is divided by $(5x + 4)$.

16 $\mathrm{f}(x) = 3x^3 - 14x^2 - 47x - 14$.

a Find the remainder when $\mathrm{f}(x)$ is divided by $(x - 3)$.

b Given that $(x + 2)$ is a factor of $\mathrm{f}(x)$, factorise $\mathrm{f}(x)$ completely.

15

16(a)

16(b)

17 a Find the remainder when $x^3 + 6x^2 + 5x - 12$ is divided by

i $x - 2$,

ii $x + 3$.

b Hence, or otherwise, find all the solutions to the equation $x^3 + 6x^2 + 5x - 12 = 0$.

a(i)

a(ii)

b



18 $\mathrm{f}(x) = 2x^3 + 3x^2 - 8x + 3$.

a Show that $\mathrm{f}(x) = (2x - 1)(ax^2 + bx + c)$ where $a, b$ and $c$ are constants to be found.

b Hence factorise $\mathrm{f}(x)$ completely.

c Write down all the real roots of the equation $\mathrm{f}(x) = 0$.

a

b

c

## Exercise 7B – Y1 Pages 142 **Luisa · 15**

19 f$(x) = 12x^3 + 5x^2 + 2x - 1$.

a Show that $(4x - 1)$ is a factor of f$(x)$ and write f$(x)$ in the form $(4x - 1)(ax^2 + bx + c)$.

b Hence, show that the equation $12x^3 + 5x^2 + 2x - 1 = 0$ has exactly 1 real solution.

a

b

## 3: The Factor Theorem **Luisa · 16**

**Factor theorem**

$(x - p)$ is a factor of a polynomial f(x) if and only if f(p)=0.

**Example** For $f(x) = x^3 + x^2 - 4x - 4$, test whether x−2 is a factor.

Example

## Examples (in textbook) — Luisa · 17

**1** Show that x−2 is a factor of $x^3 + x^2 - 4x - 4$.

1

**2: test** Factorise $f(x) = 2x^3 + x^2 - 18x - 9$ completely. First find one root.

2: test

**2: divide** Find the quadratic factor.

2: divide

**2: factorise** Factorise the quadratic.

2: factorise

Alternatively compare coefficients: in $(x-3)(ax^2+bx+c)$, a=2, c=3 and b−3a=1 gives b=7. Try integer divisors of the constant term, or use a calculator table, to find a root.

## Using Factor Theorem to find unknown coefficients — Luisa · 18

Given that 2x+1 is a factor of $6x^3 + ax^2 + 1$, find a.

**1** Choose the value making the factor zero.

1

**2** Apply the factor theorem.

2

For a linear factor ax+b, substitute x=−b/a into the polynomial.

## Test Your Understanding — Luisa · 19

$f(x) = 6x^3 + 13x^2 - 4$

**1(a)** Find the remainder when divided by 2x+3.

1(a)

**1(b)** Show that x+2 is a factor.

1(b)

**1(c)** Factorise f(x) completely.

1(c)

**2** Given that 3x−1 is a factor of $3x^3 + 11x^2 + ax + 1$, find a.

2

## Exercise 7C – Y1 Pages 146 — Luisa · 20

13. $f(x) = 3x^3 - 12x^2 + 6x - 24$.

**a** Use the factor theorem to show x−4 is a factor.

a

**b** Show that 4 is the only real root.

b

14. $f(x) = 4x^3 + 4x^2 - 11x - 6$.

**a** Show x+2 is a factor.

a

**b** Factorise f(x) completely.

b

**c** Solve f(x)=0.

c



15. $f(x) = 9x^4 - 18x^3 - x^2 + 2x$.

**a** Show that x−2 is a factor.

a

**b** Find all four real roots.

b

## Exercise Extension — Luisa · 23

**1** [MAT 2006] $x^3 + ax + b$ has factors x−1 and x−2. Find a,b.

1

**2** [MAT 2009] For which positive integers n does $x^2 - 1$ divide $n^2x^{2n+3} - 25nx^{n+1} + 150x^7$?

2

**3** [MAT 2013] $p_n(x) = (x-1) + \cdots + (x-n)$, n≥2. Find the remainder on division by $p_{n-1}(x)$.

3

Remainder theorem: division by x−a leaves remainder f(a). For any nonconstant linear divisor, evaluate f at its root.

## 4: Proof — Luisa · 24

**Conjecture**

A mathematical statement not yet proved.

Goldbach's conjecture: every even integer greater than 2 is a sum of two primes. Checking many cases supplies evidence, but does not prove every case.

**Theorem**

A mathematical statement that has been proved.

Fermat's Last Theorem: for integers n>2 there are no positive integers a,b,c satisfying $a^n + b^n = c^n$. Its proof was published in 1995.

## Types of Proof — Luisa · 25

A proof states its assumptions, follows a logical sequence, covers every case and ends with a conclusion. Deduction starts with known facts.

**1** Prove that the product of two odd integers is odd.

1

**2** Prove
$(3x+2)(x-5)(x+7) \equiv 3x^3+8x^2-101x-70.$

2

An identity holds for every permissible value: $x(x-1) \equiv x^2-x$. By contrast, $x^2=4$ holds only for x=±2.

## Be Warned... — Luisa · 26

Prove that consecutive integer side lengths of a right-angled triangle must be 3,4,5.

**Insufficient argument**

Checking $3^2+4^2=5^2$ shows one example works. It does not exclude other consecutive triples.

Proof

Start from the given assumptions, rather than assuming the conclusion you are asked to prove.

**Example** Prove $x^2 + 4x + 5$ is positive for all real x.

Example

**Your turn** Prove that the sum of the squares of consecutive odd integers is 2 more than a multiple of 8.

Your turn



**11** Prove that $x^2 + 8x + 20 \geq 4$ for all values of $x$.

11

**12** The equation $kx^2 + 5kx + 3 = 0$, where $k$ is a constant, has no real roots. Prove that $k$ satisfies the inequality $0 \leq k < \frac{12}{25}$.

12

**13** $px^2 - 5x - 6 = 0$ has two distinct real roots. Prove $p > -\frac{25}{24}$.

13

**18** $y = ax$ cuts $(x-1)^2 + y^2 = k$, k>0, at two distinct points. Prove $k > \frac{a^2}{1+a^2}$.

18

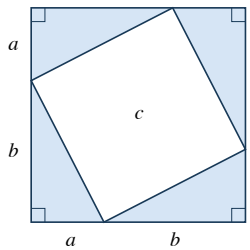


**19** Prove that $4y - 3x + 26 = 0$ is tangent to $(x+4)^2 + (y-3)^2 = 100$.

19

**20.** The diagram shows a square and four congruent right-angled triangles. Use areas to prove $a^2 + b^2 = c^2$.

Area proof



## Exercise Extension — Luisa · 31

[STEP I 2005 Q1] Prove there are 15 five-digit numbers with digit sum 43. How many have digit sum 39? Explain your reasoning.

**i** Digit sum 43

i

**ii** Digit sum 39

ii

## Other Types of Proof — Luisa · 32

**Proof by exhaustion**

Divide the possibilities into cases and prove each case.

**1** Prove $n^2 + n$ is even for every integer n.

1

**Disproof by counterexample**

One counterexample disproves a statement claiming every case works.

**2** Disprove: $n^2 - n + 41$ is prime for every integer n.

2

**4** Prove each integer cube is a multiple of 9, or one more or one less.

4

**7** Prove $(x+6)^2 \geq 2x+11$ for every real x.

7



**8** For a>0, prove $a+\frac{1}{a} \geq 2$.

8

**9(a)** For p,q>0, prove $p+q \geq \sqrt{4pq}$.

9(a)

**9(b)** Show by counterexample that the inequality need not hold when both numbers are negative.

9(b)